FAILURES IDENTIFIED

Provenance failure

At low coverage, models cite a real, on-topic paper that does not contain the specific claimed fact, rather than an invented reference.

Passes the two checks a careful reader performs: the reference exists and it is on topic.

Behavioral bias

Models infer resource scarcity from a patient's nationality and change an otherwise-invariant recommendation when the setting is unstated.

Legitimate population-specific differences were preserved (bias, not erasure).

MITIGATIONS

Verify that the cited source supports the claim

Check that the paper exists, is the record cited, and actually contains the specific claim (existence, record identity, support)

Full-text-grounded citing with abstention

Cite only from retrieved full text; abstain or seek human review when support cannot be verified or literature coverage is low

State the care setting explicitly

One clause naming a fully resourced setting removed most of the nationality-dependent variation (a low-cost prompt-level control)

Counterfactual nationality-swap audit

Re-present the same case with nationality swapped and flag any divergence before deployment

ACROSS BOTH: HOW TO EVALUATE CLINICAL LANGUAGE MODELS

Score hallucination by whether a citation supports its claim, not only whether the reference exists · stratify evaluation by literature coverage and geography

State the care setting when testing for demographic differences · build equity-focused benchmarks with, not for, underrepresented populations